

SECTION 10 – TOOLS AND TEST EQUIPMENT

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10-1 INTRODUCTION

The following tables lists the tools and test equipment needed to install the Upgrade and calibrate the system.

10-2 CRYOGENIC EQUIPMENT

TABLE 10-1
CRYOGENIC EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	2171219	RUO Temperature Monitor for LCC 1.5T & 1.0T Magnet (See Note 2)
2	46-306734G1	Low Pressure Regulator Kit (Non-magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2 & 3)
3	46-306717G1	Non-magnetic gaseous helium cylinder cart (See Note 2 & 3)
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract. 3 Equipment needed to purge coolant residue out of Gradient coil, flexible vinyl lines, and old Heat Exchanger during upgrade.		

10-3 INSTALLATION EQUIPMENT

TABLE 10-2
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinet/box from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	46-306763G1	Subminiature-D Connector Removal/Re-termination Repair Kit for Robinson-Nugent Sub-D connectors (See Note 2)
6	46-301450G1	Fiber optic connector repair kit (See Note 2)
7	46-320273G3 or G4	Non-Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non-Magnetic Tool Kits needed. May substitute both of the following kits: • 46-320273G1 Non-Magnetic Tool Kit – Metric • 46-320273G2 Non-Magnetic Tool Kit – Inch
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

10-4 TEST EQUIPMENT

TABLE 10-3
TEST EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	46-194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Tektronic 2232
2	46-194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Tektronic 2465
3	46-194427P284	Battery operated digital multi-meter, Fluke 87, 4.5 digits with frequency counter and capacitance
4	46-208572P9	Clamp-on Ammeter, 1-200A, AC/DC
5	46-317724P14	Attenuator/ load, 200 Watt, 30 dB attenuator
6	46-317724G2	RF Power Measurement Kit with Attenuator/ load, 200 Watt, 30 dB attenuator 46-317724P14
7	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
8	Catalog E6320DA or 46-194427P144	Densitometer
9	46-306801G1	Bell Gauss Meter Kit
10	46-194427P248	Microguard or ECOS leakage tester
11	46-306797G1	Fogg (ECG) Simulator and Memory Module
12	46-320433P1	Infrared Scanner 0 - 100° C
13	46-317830G1	Fiber Optic Light Meter Kit

10-5 CALIBRATION TOOLS/FIXTURES

TABLE 10-4
CALIBRATION TOOLS/FIXTURES

ITEM	GE PART NUMBER	DESCRIPTION
1	46-307500G1	Longitudinal Drive Force Gauge
2	2133388-6	System Performance Test (SPT) Phantom Set for Systems with short bore manget (See Note 1) consisting of the following: • 2125247-10 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027-2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
3	46-287900G3	Head TLT Sphere, one 100 mm Nickel Chloride sphere, and Loader (See Note 1)
4	46-287903G3	Body TLT Sphere and Loader (See Note 1)
5	46-271447G1	Full Field Distortion Phantom and case with foam
6	46-258709G1	Full Field Distortion Phantom positioner
7	46-271443G2	Head Axial Slice Analysis Phantom and case with foam incorporates the following (See Note 2): • Transmit/receive Quad Phase • Slice Thickness • Slice Offset • Resolution
8	46-271444G1	Sagittal/Coronal Slice Analysis Phantom and case with foam (See Note 2)
9	46-255816G1	RF Test Cables Kit (See Note 2)
10	46-307164G4	1.5T / 1.0T GRAFIDY II Kit (See Note 2)
11	46-307164G6	1.0T GRAFIDY II Kit (See Note 2)
12	46-301549G1	TPS RF Test Cable/Adapter Kit (See Note 2)
13	46-301927G1	TPS RF Service Interface Kit (See Note 2)
14	46-306712G1	Torque driver kit (See Note 2)
15	2101360	Power Supply Calibration Kit (See Note 2)
16	46-320006P1	DAT Cleaning Kit (See Note 1, shipped with system software)
17	46-265826G3	Head Signal-to-noise (SNR) sphere (See Note 3)
18	46-265635G4	Body Signal-to-noise (SNR) sphere (See Note 3)
19	46-265635G2	Body T2 sphere (See Note 3)
20	46-328480G1	100 mm Nickel Chloride sphere and Universal Phantom Holder (See Note 2)
Note	1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract. 3 Customer may purchase these items.	

10-6 TOOL KIT

TABLE 10-5
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Extension cords, with ground conductor
2	—	Power strip, grounded type, with minimum of five outlets
3	—	Soldering iron, pencil type with solder
4	—	Solder sucker
5	—	Assortment of Brady Quick labels
6	—	Micro clip leads
7	—	14 pin and 16 pin DIP clips
8	46-258218P3	Vinyl electrical tape
9	46-258218P4	Copper tape, 3 in. wide
10	46-258218P5	Copper tape, 2 in. wide
11	46-258218P6	Copper tape, 1 in. wide
12	—	Alcohol cleaning solution
13	—	Plastic or aluminum flashlight
14	—	Plastic or aluminum pen light
15	AMP No. 458994-1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183-R	Pin extractor, M-series
17	46-237072P1	Pin extractor, Sub-D
18	46-307307G1	Crimping tool for coax cable and BNC connectors Inserts for RG8, 9, 11, 214 46-255841P103 RG58, 223 46-255841P100 RG59 46-255841P101 RG174 46-255841P102 Die Removal Tool 46-255841P201
19	—	Assorted crimp tools.
20	—	Non-magnetic level
21	—	Non-magnetic tape rule, 12 ft
22	—	Assorted drill bits
23	—	Inspection mirror
24	—	Tap set, standard, and tap handle, T-type
25	—	6 in. rule, standard and metric markings
26	—	Alignment tool (tweaker)
27	—	Hex/alignment tool
28	—	Hemostat, 5 in., curved
29	46-198094P1	Wrist grounding strap
(Continued)		

10-8 TOOL KIT (Continued)

TABLE 10-7 (Continued)
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
30	46-400123P72	IC Extractor Tool
31	Xcelite 110CG	Diagonal Cutting Pliers, 4-1/2 in.
32	—	Screw Starter, aluminum or plastic shaft
33	—	Hobby and utility knives
34	—	Spring scales, 0-10 lbs and 0-50 lbs
35	46-287192P1	Extractor for EIMAC 3CX 800, 8877, 8874 Tubes
36	46-313413P1	Extractor for 20 - 100 pin PLCC Chips